

A New Time Series Framework for Forest Fire Risk Forecasting and Classification

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Abstract—There's an increasing concern about the occurrence and spread of forest fires across the globe, as they contribute to greenhouse gas emissions and play a major influential role in economics and public health. Thus, there's a need for accurate methods to predict and classify forest fire risk. The main known forest fire risk indexes have limitations, such as not taking into account the unique characteristics of the biome in study, and not being able to predict forest fire risk for a given number of days in the future. This last aspect, in particular, is of utmost relevance. Addressing it allows for coordinated planning and action by proper authorities with adequate anticipation. Aiming to solve this problem, we present a new framework that applies Machine Learning methods for: (1) climatic variables forecasting; and (2) forest fire risk classification. For the first objective, different time series forecasting algorithms were tested. The forecasted variables are then used as input for the second objective, for which different classification algorithms were also tested. We evaluated our proposal using Brazilian Pantanal regional biome data from 1999 to 2019, where climatic variables were collected from ground meteorological stations, and fire occurrences (hotspots) were obtained from satellite images. The experiments considered 4 climatic variables and 5 forest fire risk classes. The results were evaluated based on the average correlation between (i) the prediction of forest fire risk classes and (ii) the observation of hotspots. Our proposal proved to be better or competitive with the main forest fire risk indexes, with the advantage of predicting fire risk for a given number of days in the future.

Index Terms—forest fire risk, supervised machine learning, classification, time series forecasting

I. INTRODUCTION

Forest fires are becoming a great challenge due to changes in both climate and land usage dynamics. They contribute to greenhouse gas emissions and play a major influential role in economics and public health. Between 2001 and 2019, forest fires were the cause of one-third of global deforestation. Particularly within the tropical primary forests of Africa and

Latin America, the worldwide area of fire-related deforestation is increasing [1].

Government entities from different countries use monitoring and prediction systems for forest fire prevention and fire-fighting planning. Through the use of meteorological data, it is possible to classify the risk of forest fire occurrence and, by doing so, contribute to forest fire prevention and combat. Different studies, presented in Section II-B, apply regular forest fire risk indexes for this purpose. However, the application of such indexes has limitations, such as: (1) not being able to predict forest fire risk for a given number of days in the future; (2) not being adjusted according to the characteristics of each biome; and (3) being limited to specific climatic variables.

Besides the regular forest fire risk indexes, models based on Machine Learning (ML) can also be used for forest fire risk classification. These have the advantage of taking into account the characteristics of the region under study and not limiting the amount or type of variables that can be used to describe the region. Recent studies have shown that the use of Artificial Neural Networks (ANN), a class of ML algorithms, can lead to great results for the prediction of forest fires [2]–[4]. However, none of these studies presented a model capable of predicting forest fire risks for a certain number of days in the future, which would help plan coordinated actions by authorities.

In this sense, it is possible to use the knowledge linked to the area of time series forecasting. A time series consists of data that concerns the observation of a phenomenon over time, as is the case with climatic variables. To predict time series, statistical methods based on autocorrelations present in the data can be applied, as well as models based on ML.

Regarding the Brazilian Pantanal region specifically, there is a continuous effort of the environmental authorities to forecast and combat forest fires. An example is the SARIPAN framework [5]. However, SARIPAN can only make predictions for the same day the climatic variables are collected - i.e., it

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is not able to forecast the forest fire risk for a number of days in the future. Besides that, it does not make use of ML, which means that it cannot learn from historical data neither adapt to environmental changes in the climate. Instead, it applies the 5 forest risk rate indexes that are described in Section III-B. These same indexes are used as baselines in the comparisons to our proposed framework.

To address the above mentioned limitations, we propose a new framework that applies ML for climate data forecasting and forest fire risk classification in Pantanal. It combines time series forecasting and classification algorithms, trained using climate and hotspot data. The proposal was evaluated using Brazilian Pantanal regional biome data from 1999 to 2019, collected from ground meteorological stations and satellite images. The time series forecasting algorithms investigated were Long Short-Term Memory (LSTM) [6], Gated Recurrent Unit (GRU) [7], and Convolutional Neural Network (CNN) [8]. The classification algorithms applied were Decision Tree [9], Random Forest [10], XGBoost [11], LightGBM [12], CatBoost [13], Naive Bayes [14], K-Nearest Neighbor (KNN) [15], Multi-Layer Perceptron (MLP) [16] and Support Vector Machine (SVM) [17]. The results were compared to the most applied forest fire risk indexes known in the literature for Pantanal. For the overall results, our framework outperformed all indexes regarding the “High” and “Very High” fire risk classes, which are the most important ones for the anticipation of actions by proper authorities. When considering the lower levels of fire risk classes, the framework is competitive with the other risk indexes, while outperforming them for “Null” class. In addition, for all fire risk classes, our proposal has the advantage of making predictions for up to 3 days in the future.

II. RELATED WORK

A. Time Series Forecasting

A time series consists of data concerning the observation of a phenomenon over time. Algorithms for time series forecasting can act in different ways, but the main ones are able to identify patterns and learn from historical data. Once trained, they can predict data for a number of future intervals [18] - i.e., forecast data for a given number of days in the future (“forecast horizon”), based on a number of days in the past (“observation window”).

Since they were proposed in 1997, LSTMs have been applied in different areas to solve a wide variety of problems, including time series forecasting. As an example, LSTMs were studied in the context of energy consumption forecasting [19]. Fourteen years of energy consumption data collected hourly were analyzed and it was indicated that LSTMs can increase the accuracy of energy consumption forecasting, when compared to other forecasting algorithms. In [20], a LSTM model was built with seasonal wind speed data in 4 different regions of India. The results showed that seasonal wind speed forecasting using the LSTM model could reduce the errors of conventional weather forecasting.

Another ANN algorithm that is often applied in time series forecasting is GRU. It was applied in conjunction with a factoring machine (FM-GRU) for forecasting water quality over time [21], and applied to predict machinery failures obtaining 87% accuracy [22].

Although CNNs are mainly used in image classification and natural language processing tasks, they have also been applied to time series forecasting. As an example, CNNs were applied for stock price time series forecasting, showing better performance when compared to LSTMs [23]. CNNs were used for forecasting electric and solar power time series [24]. The results were compared with LSTMs and MLPs, and CNNs were shown to be competitive with MLPs, while showing better results than LSTMs.

B. Forest Fire Risk Prediction

The study of [25] evaluated different forest fire risk indexes - Angström, Telicyn, Nesterov, Monte Alegre Formula (FMA) and FMA⁺ (such indexes will be detailed in Section III-B) - for the Brazilian Pantanal. They were compared according to their accuracies on the classification of forest fire risk versus identified hotspots. The FMA excelled over all the others.

A MLP was used to predict the number of forest fires and to predict the burned area of the Brazilian Pantanal region [26]. Secondary meteorological data and satellite images were used. The method obtained an accuracy of 99.4% using a time series dataset to predict the number of fire outbreaks. It was also possible to obtain an accuracy of 98.6% using the same dataset for prediction of burned area. The results suggest that employing ANNs in Pantanal meteorological data leads to well accurate results.

Another ANN was proposed to predict burned area of forest fires in the northwestern region of Portugal [3]. Data from climatological stations with spatial and temporal components were used, as well as 4 different weather conditions. The model achieved an accuracy of 98.87%.

However, the above mentioned studies have limitations: They do not predict fire risk for a number of days in the future [3], [25], [26], and cannot be adjusted according to the characteristics of each biome [25]. In addition, they are limited to specific climatic variables [25].

III. MATERIALS AND METHODS

A. Datasets

The Pantanal region biome was chosen as the focus of this study due to its ecological importance. It is a wetland area of about 150,355 km², with most of its territory belonging to Brazil, but also touching Bolivia and Paraguay. The Pantanal region biome is a Biosphere Reserve, recognized by UNESCO, and, although it still maintains approximately 80% of its vegetation cover [27], the ecosystems of this biome are constantly threatened by anthropogenic activities, but also by a complex seasonal dynamics of its climate. Still, the framework proposed in this study (Section IV), is fully configurable and can train models given the climate and hotspot data from any region.

For training forecasting and classification models, 2 datasets are required to our framework: climate and hotspot. The climate data is the one which contains the climatic variables for a given region (attributes). The hotspot data is the one which contains the days when hotspots (fire) were observed for that same region (target).

1) *Climate Data*: The climate dataset consists of a 21-year time series (1999-2019) of air temperature ($^{\circ}C$), relative humidity (%), wind speed (m/s) and daily precipitation (mm). The temperature, relative humidity and wind speed data are measured daily at 1 pm (official Brazilian time, UTC-03). The precipitation data, on the other hand, is regarding the daily accumulated value. In total, there are 7670 instances. Data were collected from ground meteorological stations at the Main Climatological Station of Nhumirim city (latitude $18^{\circ}59'21''S$, longitude $56^{\circ}37'25''W$, altitude 102 meters), located in the sub-region of Nhecolândia city in the Brazilian Pantanal [25]. Table I summarizes the climate dataset, where ‘T’ stands for ‘Temperature’, ‘RH’ for ‘Relative Humidity’, ‘P’ for ‘Precipitation’, and ‘WS’ for ‘Wind Speed’.

TABLE I
DESCRIPTION OF THE CLIMATE DATASET USED IN THE STUDY.

	Date	T	RH	P	WS
Type	Date	Numeric	Numeric	Numeric	Numeric
Measure	-	$^{\circ}C$	%	mm	$m \cdot s^{-1}$
Unique	7670	257	109	344	14
Min.	01/01/1999	9.00	14.00	0.00	0.00
Avg.	-	30.95	58.59	3.03	3.46
25%	-	28.70	47.00	0.00	2.00
50%	-	32.00	57.00	0.00	3.00
75%	-	34.10	70.00	0.00	4.00
Max.	31/12/2019	40.70	100.00	203.60	20.00

2) *Hotspot Data*: The hotspot dataset covers the same period as the climate dataset. There are 15499 instances and the information was collected from the database provided by the Image Processing Division of the National Institute for Space Research (INPE) [28]. It was generated by the NOAA12 (until 2002) and AQUA_M-T (from 2002) satellites. Hotspots are detectable through spatial resolution elements (pixels) that highlight high temperatures. These present the lowest gray level values in the images of the thermal infrared region band 3 ($3.7\mu m$) of the AVHRR radiometer sensor. This band measures radiant energy emission from the Earth’s surface, in which saturated pixels correspond to a temperature of at least $47^{\circ}C$, normally associated with burning targets [25]. The data were filtered to eliminate the spots that were outside of the related coverage area, that is, a radius greater than 100 km and beyond the vertical line near the hills of the Nhumirim city station. Table II presents a snapshot of the mentioned data.

B. Forest Fire Risk Indexes

1) *FMA*: Monte Alegre Formula (FMA) is the main forest fire risk index developed based on Brazilian climatic conditions [29]. It is based on the central region of Paraná state known as Monte Alegre. Its variables are relative humidity, which must be measured at 1 pm, and precipitation. It is a

TABLE II
SNAPSHOT OF THE HOTSPOT DATASET USED IN THE STUDY. THE INSTANCES REPRESENT DAYS IN WHICH A HOTSPOT WAS DETECTED.

Date	Satellite	City	Latitude	Longitude
22/01/1999	NOAA-12	Corumbá	-19.1767	-55.9281
01/06/1999	NOAA-12	Corumbá	-18.625	-56.0417
...	...	...	...	...
15/12/2019	AQUA_M-T	Corumbá	-19.118	-55.945
28/12/2019	AQUA_M-T	Aquidauana	-19.369	-55.577

cumulative index, that is, it depends on the daily measurement of these climatic variables. The longer the sequence of days with low relative humidity and no rain, the greater the risk of forest fire. FMA formula is described in Equation 1, where n is the number of days without rain (precipitation less than 13 millimeters), and H is the relative humidity (%) measured at 1 pm.

$$FMA = \sum_{i=1}^n \frac{100}{H_i}, \quad (1)$$

2) *FMA⁺*: An updated version of FMA [30], which includes the wind speed variable in the index calculation. The main objective was to create an index that also identifies the potential for fire propagation and not just its ignition. FMA⁺ is also a cumulative index, although the wind speed variable is not - the wind speed value at 1 pm each day is used. The index can be calculated according to Equation 2, where n is the number of days without rain (precipitation less than 13 millimeters), H is the relative humidity (%) measured at 1 pm, and v is the wind speed (m/s) measured at 1 pm.

$$FMA^+ = \sum_{i=1}^n \left(\frac{100}{H_i} \right) \cdot e^{0.04 \cdot v_i}, \quad (2)$$

3) *Telicyn*: The Telicyn logarithmic index (I) [31], given in Equation 3, where T is the air temperature ($^{\circ}C$) at 1 pm, and r is the dew point temperature, i.e., the temperature at which the water vapor present in the air ($^{\circ}C$) changes to a liquid state via condensation. The dew point temperature (r) can be obtained from the correlation between air temperature and relative humidity [32].

$$I = \sum_{i=1}^n \log(T_i - r_i), \quad (3)$$

4) *Angström*: The fire hazard index, or Angström formula (B). Developed in Sweden and widely used in parts of Scandinavia [34]. It is calculated by Equation 4, where H is the relative humidity (%) measured at 1 pm; and T is the air temperature ($^{\circ}C$) at 1 pm. This index does not require any specific adjustment and is not a cumulative index.

$$B = \left(\frac{H}{20} \right) + \left(\frac{T - 27}{10} \right), \quad (4)$$

5) *Nesterov*: The Nesterov index [35] focuses on the so-called degree of flammability, which provides a sense of the danger of fires. It is based on the sum of ‘dangerous’ days, i.e., days whose saturation deficit is high. Equation 5

presents the calculation of the Nesterov index (G), where n is the number of days without rain, d is the saturation deficit (millibars) at 1 pm, and T is the air temperature ($^{\circ}C$) at 1 pm. The saturation deficit can be calculated from a correlation of relative humidity, air temperature and maximum pressure of water vapor [32].

$$G = \sum_{i=1}^n d_i \cdot T_i, \quad (5)$$

The values obtained by Equations 1, 2, 3, 4 and 5 must be converted to one of the 4 forest fire risk classes, which can be obtained from Table III.

TABLE III
FMA, FMA⁺, TELICYN (I), ANGSTRÖM (B) AND NESTEROV (G) FOREST FIRE RISK CLASSES [32], [33].

Forest Fire Risk Class	FMA	FMA ⁺	I	B	G
Null	≤ 1.0	≤ 3.0	≤ 2.0	> 4.5	≤ 300
Low	1.1 to 3.0	3.1 to 8.0	2.1 to 3.5	4.3 to 4.5	301 to 500
Medium	3.1 to 8.0	8.1 to 14.0	3.6 to 5.0	4.0 to 4.2	501 to 1000
High	8.1 to 20.0	14.1 to 24.0	> 5.0	3.5 to 3.9	1001 to 4000
Very High	> 20.0	> 24.0	-	< 3.5	> 4000

C. Evaluation Measure

We evaluated the results using the correlation between the prediction of each forest fire risk class and the occurrence of a hotspot. This correlation is given by Equation 6, where $corr_C$ is a value between 0 and 1, representing the correlation value for a class C , P_C is the set of days on which class C was predicted, and I is the set of days on which one or more hotspots were observed.

$$corr_C = \frac{|P_C \cap I|}{|P_C|}, \quad (6)$$

It is fair to assume that the lowest the correlation for classes “Null”, “Low” and “Medium”, the best. As well as it is desired to have the highest possible correlation for classes “High” and “Very High”.

The review of evaluation measures applied to forest fire risk [25], [36], [37] shows there is no consensus on the use of a single evaluation measure. Also, it is not possible to apply the conventional Machine Learning evaluation measures, such as Accuracy and F-Measure, to evaluate our framework performance. This is because these measures rely on knowing the true values of the instances. In our study, however, the predictions are the forest fire risk classes, while the known true values are the occurrence of hotspots.

Given the described scenario and the context of this study, the correlation presented in Equation 6 was chosen as the evaluation measure, specially due to its simplicity and interpretability.

IV. FRAMEWORK

Our proposed framework¹ was developed using the Python 3 programming language, and is capable of running for a large set of framework hyperparameters: (i) Scaling method; (ii) Forecasting algorithm; (iii) Classification algorithm; (iv)-(vii) Forest fire risk class thresholds (thr_{null} , thr_{low} , thr_{medium} and thr_{high}). There is no “Very High” forest fire risk class threshold, because this corresponds to $1 - thr_{high}$, as will be explained later in Section IV-A.

The framework iterates over different combinations of these hyperparameters. This is summarized in Figure 1.

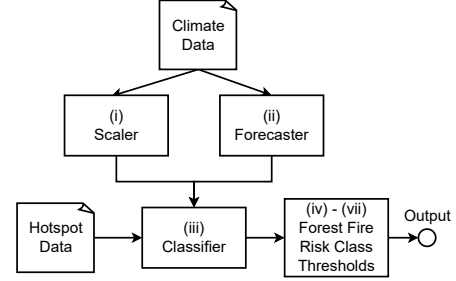


Fig. 1. Summary each framework iteration, which will run for a combination of (i) to (vii) framework hyperparameters.

The forecaster in Figure 1 is responsible for predicting climatic variables for a given number of days in the future. The classifier predicts whether there has been and/or will be a hotspot in a given day, as per its climatic variables. The implemented framework is composed of 2 different pipelines: “Training” and “Prediction”, that are described in the following sections. Ideally, “Training” pipeline will be executed only once, or every time one wants to retrain the model. The “Prediction” pipeline, on the other hand, should be run for every forest fire risk prediction.

A. “Training” pipeline

This pipeline trains a set of scaling methods, time series forecasting and supervised classification models, and selects the best combination. Its first step is data preparation, i.e., the climate and hotspot datasets are combined. The new dataset contains the climatic variables and a binary column that indicates whether a hotspot was identified (1) or not (0) in the corresponding day (Table IV). There are 5287 instances where no hotspot was identified (68.93%), against 2383 where it was (31.07%).

The second step splits data into 80% training plus validation ($X_{training_validation}$) and 20% test (X_{test}). The first is again splitted into 80% training ($X_{training}$) and 20% validation ($X_{validation}$), which leads to a split of 64%/16%/20%. The data is not shuffled, keeping the historical information ordered to be used by the forecaster later on. We fit the scaling method on $X_{training}$, and the scaled data is used train the multivariate forecaster. The goal is to learn how to predict

¹Code available at <https://github.com/bzamith/IJCNN2023PantanalFireDetection>

TABLE IV
SNAPSHOT OF THE DATASET CREATED IN THE FIRST STEP OF THE
“TRAINING” PIPELINE.

Date	T	RH	R	WS	hotspot
01/01/1999	29.7	79.0	2.0	0.0	0
01/02/1999	30.7	69.0	3.0	4.0	0
...	...	...	...	...	...
12/30/2019	36.9	50.0	1.0	0.0	1
12/31/2019	26.9	80.0	2.0	5.8	0

climatic variables for the next N days (forecast horizon), based on the past W days (observation window). Thus, the forecaster generates N new datasets, with N being equal to the number of days in the future for which one wish to predict the forest fire risk.

As mentioned earlier, there are different scaling methods and forecasting algorithms that can be tested. Hence, the framework will test all possible combinations, and select the best one based on the lowest Mean Absolute Error (MAE) obtained on $X_{validation}$. The MAE, when compared to other often-used error measures, is a more natural measure of average error and is unambiguous [38]. It is recommended for dimensioned evaluations and inter-comparisons of average model-performance errors, which is our case.

After choosing the scaling method and forecasting algorithm, the classification algorithm is selected. To do so, different algorithms are trained with $X_{training}$. During this process, each algorithm goes through hyperparameter optimization using 20% of $X_{training}$ as validation. The best classification model is chosen based on the highest F-Score obtained on $X_{validation}$.

Since the outputs of the classifiers can be interpreted as the probability of occurring a hotspot, thresholds are needed to convert these outputs to one of the 5 forest fire risk classes. Upon applying the forest fire risk rate class thresholds, the probability for a given instance i is converted to a class C_i (Equation 7).

$$C_i = \{\text{If } prob_i < thr_{null}, i \in \text{Null or} \\ thr_{null} \leq prob_i < thr_{low}, i \in \text{Low or} \\ thr_{low} \leq prob_i < thr_{medium}, i \in \text{Medium or} \\ thr_{medium} \leq prob_i < thr_{high}, i \in \text{High else} \\ i \in \text{Very High}\}, \quad (7)$$

Then, the correlation measure (Section III-C) is calculated for each class (“Null”, “Low”, “Medium”, “High” and “Very High”). This is performed for each day in the present or future ($0 \leq n \leq N$) on $X_{validation}$. To select the best combination of thresholds, a score $S = \frac{S_1 + S_2}{15}$ is calculated, with S_1 and S_2 given by Equations 8 and 9, where $corr_{n,c}$ is the correlation metric for a given class c in day n . The combination that provides the highest score S is chosen as the selected thresholds.

$$S_1 = \sum_{n=0}^N 5 \cdot (1 - corr_{n,null}) + 4 \cdot (1 - corr_{n,low}) + 3 \cdot (1 - corr_{n,medium}) \quad (8)$$

$$S_2 = \sum_{n=0}^N 2 \cdot corr_{n,high} + 1 \cdot corr_{n,very_high} \quad (9)$$

As can be seen in Equation 8, we considered the difference $1 - corr_{n,class}$ for classes “Null”, “Low” and “Medium”. As explained in Section III-C, $corr_C$ measures the correlation between the prediction of class C and the occurrence of a hotspot. We thus expect the lowest correlation possible, since there should be no detected hotspots when classes “Null”, “Low” and “Medium” are predicted. The opposite is true for classes “High” and “Very High”.

We can also see in Equation 8 that the correlations for “Null” and “Low” are multiplied by higher factors in the weighted average. Thus, they have a higher contribution to the final score S . This is because in our application, false negatives are a bigger concern compared to false positives. False negatives happen when the model predicts there will be no forest fire in a given day (“Null” and “Low”), but the forest fire occurs. False positives occur when the model predicts there will be forest fire in a given day (“High” and “Very High”), but the forest fire doesn’t occur. It is thus clear that false negatives are a bigger issue, since by predicting there will be no forest fire in a given day, the proper authorities may not be prepared to act upon the forest fire occurrence. This justifies our decision on weighting the correlations in a decreasing manner.

Once the best thresholds are defined, based on the highest score S for $X_{validation}$, the selected scaling, forecasting and classification models are re-trained with $X_{training} + X_{validation}$ (80%) and applied to X_{test} (20%). The results are then used to report the final performance of the framework.

B. “Prediction” pipeline

This pipeline predicts the fire risk for a forecast horizon (N), based on an observation window (W). To do so, it uses the models (scaling, forecasting and classification) trained and selected by the previous pipeline, and receives as input a dataset with climatic variables measured on the previous W days. The output is the forecasted climatic variables for the next N days and their respective forest fire risk classes. Depending on the application, this pipeline should ideally be run on daily basis, by simply keeping the prediction dataset up to date and then triggering the pipeline execution.

V. EXPERIMENTS

We evaluated our framework regarding its performance in predicting forest fire risk for up to 3 days in the future ($N = 3$). To do so, the forecasting algorithm is configured to make predictions based on a past 5 days window ($W = 5$). These numbers are arbitrarily chosen, and other values could be tested as well. Given that currently in Pantanal there are no methods to predict future forest fire risks, we consider that being able predict 3 days in the future is already an important innovation, at the same time allowing the proper authorities to become prepared and take actions when necessary.

A. Scaling Methods

We considered a space of 5 scaling methods: Min-Max (scales each feature such that their values are between 0 and 1), Standard (scales features by removing the mean and scaling to unit variance), Robust (scales features by removing the median and scaling them according to the interquartile range), Max-Abs (scales such that the maximal absolute value of each feature is 1), and None (no scaling method is applied to the features).

B. Forecasting Algorithm

We evaluated 3 forecasting algorithms: Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), and Convolutional Neural Network (CNN). The implemented Long Short-Term Memory makes use of 2 LSTM layers with 100 output neurons each, and 1 Dense layer that applies the sigmoid activation function. While the implemented Gated Recurrent Unit architecture makes use of 2 GRU layers, also with 100 output neurons each, and 1 Dense layer that applies the sigmoid activation function. The implemented CNN was adapted from [39]. It uses a raw CNN layer for feature extraction (64 filters with relu), combined with LSTMs to support sequence prediction. This type of architecture is often referred as CNN-LSTM.

All forecasting algorithms were trained for up to 250 epochs, stopping if no error improvements were detected after 10 epochs.

C. Classification Algorithm

The classifiers evaluated were Decision Trees, Random Forests, XGBoost, LightGBM, CatBoost, Naive Bayes, K-Nearest Neighbors (KNN), Multi-Layer Perceptron (MLP) and Support Vector Machines (SVM). The forecasting and classification algorithms were chosen based on our literature review of recent state-of-the-art works.

The classification algorithms were trained using a cross-validation with 3 folds to select the classifier hyperparameters. We chose a low number of fold in order to keep a relevant size of training data. For the MLP and SVM algorithms, a limit of 200 epochs was applied.

D. Forest Fire Risk Class Thresholds

In order to find suitable thresholds for each fire risk class, we evaluated all possible combinations of values, ranging from 0 to 1, with increasing steps of 0.05 (21 values in total). However, for a combination of thresholds to be valid, we considered that $thr_{null} < thr_{low} < thr_{medium} < thr_{high}$ so that we satisfy Equation 7.

VI. RESULTS

A complete experiment to find the best hyperparameters takes into account 5 scaling methods and 3 forecasting algorithms ($5 \times 3 = 15$ combinations), 9 classification algorithms and 194,481 possible threshold combinations ($21 \times 21 \times 21 \times 21$). This results in 194,505 iterations ($15 + 9 + 194,481$).

TABLE V
COMBINATION OF HYPERPARAMETERS SELECTED BY THE FRAMEWORK FOR THE EXPERIMENTS (SECTION V).

Parameter	Value
Scaling Method	Max-Abs
Forecasting Algorithm	Long Short-Term Memory
Classification Algorithm	CatBoost
thr_{null}	0.05
thr_{low}	0.1
thr_{medium}	0.25
thr_{high}	0.65

The final combination of hyperparameters selected by the framework is described in Table V.

The Catboost classifier hyperparameters selected via cross-validation were: learning rate equal to 0.04; depth equal to 6, and leaf estimation method equal to “Gradient”.

We calculated the correlations between our predicted forest fire risk classes and the occurrence of hotspots, and compared to the correlations provided by FMA, FMA⁺, Telicyn, Angström and Nesterov forest fire risk indexes. The experimental results were obtained with 3 different stages: (i) First, the forecaster is applied to the 4 climatic variables; (ii) then, the forest fire risk indexes are applied to the present and forecasted climatic variables; (ii) finally, the classifier is applied for the present and forecasted climatic variables. Table VI presents our experimental results, with the best results highlighted in bold face; “Fram.” stands for “Framework”, “Angs.” stands for “Angström” and “Nest.” stands for “Nesterov”. In the same way, Table VII presents the number of predictions for each class.

TABLE VI
CORRELATIONS BETWEEN EACH CLASS AND METHOD.

	Fram.	FMA	FMA ⁺	Telicyn	Angs.	Nest.
Present						
Null	0.03	0.06	0.09	0.15	0.19	0.06
Low	0.11	0.13	0.21	0.32	0.12	0.18
Medium	0.21	0.25	0.33	0.35	0.12	0.13
High	0.46	0.32	0.28	0.48	0.16	0.30
Very High	0.80	0.52	0.48	-	0.46	0.47
1 Day in the Future						
Null	0.40	0.00	0.00	0.50	0.00	0.00
Low	0.17	0.50	1.00	1.00	0.50	0.00
Medium	0.19	1.00	1.00	1.00	0.19	0.50
High	0.46	0.43	0.67	0.33	0.17	1.00
Very High	0.66	0.33	0.33	-	0.45	0.33
2 Days in the Future						
Null	0.00	0.00	0.00	0.50	0.00	0.00
Low	0.24	0.50	1.00	1.00	0.33	0.00
Medium	0.21	1.00	1.00	1.00	0.34	0.50
High	0.46	0.29	0.50	0.33	0.20	1.00
Very High	0.64	0.33	0.33	-	0.44	0.33
3 Days in the Future						
Null	1.00	0.00	0.00	0.50	0.00	0.00
Low	0.38	0.50	1.00	1.00	0.00	0.00
Medium	0.23	1.00	1.00	1.00	0.42	0.50
High	0.47	0.25	0.33	0.33	0.23	1.00
Very High	0.55	0.33	0.33	-	0.43	0.33

TABLE VII
NUMBER OF PREDICTIONS FOR EACH CLASS AND METHOD.

	Fram.	FMA	FMA ⁺	Telicyn	Angs.	Nest.
Present						
Null	207	148	278	564	78	212
Low	195	171	239	153	60	55
Medium	363	301	156	121	110	110
High	620	365	163	696	395	399
Very High	149	549	698	-	891	758
1 Day in the Future						
Null	5	0	1	2	0	0
Low	92	2	2	2	8	0
Medium	732	2	1	1	64	2
High	596	7	3	1529	589	2
Very High	109	1523	1527	-	873	1530
2 Days in the Future						
Null	1	0	1	2	0	0
Low	87	2	2	2	6	0
Medium	792	3	2	1	61	2
High	564	7	2	1529	667	4
Very High	90	1522	1527	-	800	1528
3 Days in the Future						
Null	1	0	1	2	0	0
Low	34	2	3	2	0	0
Medium	919	3	1	1	19	2
High	486	8	3	1529	775	4
Very High	94	1521	1526	-	740	1528

As can be seen from Table VI, our forecasting feature is able to provide good results for all forest fire risk classification methods (framework and indexes). Regarding our implemented classification model specifically, it outperforms all forest fire risk indexes regarding the “Very High” class. For the present forest fire risk, it also outperforms the indexes for “Null” and “Low” classes.

Besides it, Tables VI and VII show that, as the number of days in the future grows, there is a tendency of predicting higher forest fire risk classes. This is expected, as the error derived from the forecaster ends up propagating to the classifier and forest fire risk indexes, hence to the final correlations. But most importantly, the correlations for both the framework and the forest fire risk indexes do not degrade significantly as the number of days in the future grows.

In order to perform an overall evaluation considering all days, we recalculated the correlations by applying a weighted average across each class, and summing up the days (present and future). Equation 10 presents the calculations, where $count_{n,C}$ is the number of predictions for class C in the n ’th day. Table VIII presents the results.

$$weighted_corr_C = \sum_{n=0}^N \frac{corr_{n,C} \cdot count_{n,C}}{count_C} \quad (10)$$

The overall results, from Table VIII, show that our framework overcomes the limitation of most proposals known in the literature, which cannot predict forest fire risk for a number of days in the future. For the framework’s classification feature, it is possible to note from Table VIII that it outperforms all forest fire risk indexes regarding the “High” and “Very High” classes. When considering the lower levels of classes, such as “Null”,

TABLE VIII
FINAL WEIGHTED CORRELATION FOR EACH CLASS AND METHOD.

	Fram.	FMA	FMA ⁺	Telicyn	Angs.	Nest.
Null	0.04	0.06	0.09	0.15	0.19	0.06
Low	0.18	0.15	0.24	0.35	0.18	0.18
Medium	0.21	0.27	0.35	0.36	0.21	0.15
High	0.46	0.32	0.29	0.35	0.20	0.32
Very High	0.68	0.35	0.35	-	0.45	0.35

“Low” and “Medium”, the framework is competitive with FMA and Nesterov forest fire risk indexes. It is a significant result, given that false negatives are a bigger concern than false positives, within the context of the application, as exposed earlier in Section IV-A.

The experiments were run in 3 different hardware configurations, summarized in Table IX. The results presented in Tables V to VIII are based on the execution in Machine 1. Across all machines, the “Training” pipeline took approximately 2 hours and 30 minutes on average to execute, while the “Prediction” pipeline took an average of 30 seconds to conclude. Ideally, the first pipeline can be executed only once or every time one wants to retrain the model. Only the “Prediction” pipeline needs to be run frequently.

TABLE IX
TECHNICAL SPECIFICATIONS OF THE MACHINES USED.

	Machine 1	Machine 2	Machine 3
OS	macOS Catalina	Ubuntu	Windows 10
Processor	Intel Core i5 @ 2.30 GHz	Intel Core i7 @ 2.80GHz	Intel Core i5 @ 1.70GHz
RAM	8 Gb	16 Gb	16 Gb
GPU	Intel Iris Plus Graphics 640	NVIDIA GeForce GTX 1050	Intel UHD Graphics

VII. CONCLUSION

This paper proposed a framework which applies ML for climate data forecasting and forest fire risk classification. Extensive experiments with data from the Brazilian’s Pantanal region showed that, when compared to state-of-the-art baseline forest fire risk indexes, our framework significantly improved the ability to classify “High” and “Very High” risks of forest fire, while being competitive for “Null”, “Low” and “Medium” classes.

Our framework can be executed for any biome and territorial region, provided that reliable climate and hotspot data are available. To the best of our knowledge, there is currently no research concerning the development of a framework with capacity to jointly forecast climate data and classify the forest fire risk for both the present and the future.

For the Pantanal region specifically, there is a continuous effort of the environmental authorities to forecast and combat forest fires. However, the existing frameworks that classify forest fire risk, such as SARIPAN [5], are limited to the current day - that is, are not able to forecast the forest fire risk for a number of days in the future. Thus, the proposed framework overcomes this limitation as it is capable of forecasting the

forest fire risk for a number of days in the future. The experimental results indicate that the performance was competitive or even better than the forest fire risk indexes, and is kept at high level even for a 3 days forecast. In fact, the indexes used as baseline for comparison are the same as the ones applied by SARIPAN, hence demonstrating the importance and potential impact of the proposed framework.

As future work, we intend to investigate AutoML for climatic variables prediction and forest fire risk classification. AutoML has the potential to improve the results given its ability to efficiently search in a huge hyperparameters space.

VIII.

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